HCC1395 | EIF3K-CYP39A1 fusion

531-aa complete coding candidate - RNA reconstruction, not measured translation

149 exact Illumina junction fragments

9 ONT coding-span reads; 4 also have the exact junction

The reference-guided 1,596-nt coding RNA path is tiled by exact original reads, including 19 mapped Illumina reads with overlaps of at least 20 nt. The Illumina input is fusion-enriched. Counts do not measure whole-library prevalence.

IGV (GRCh37): chr19:39123318-39123318 chr6:46607405-46607405

Protein architecture and reference alignment

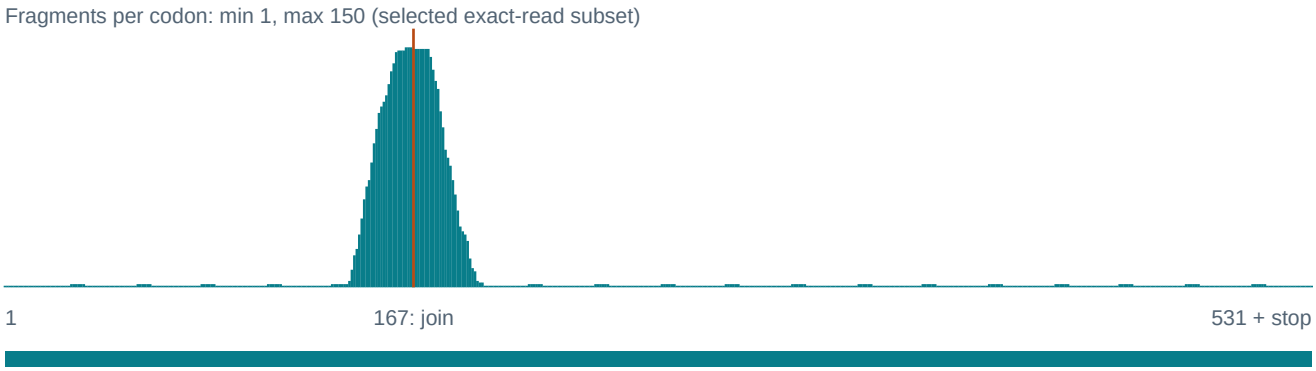
Fusion residues 1-166 match EIF3K residues 1-166. Residue 167 is the split junction codon: one EIF3K base plus two CYP39A1 bases. Its alanine matches CYP39A1 residue 105. Fusion residues 168-531 match CYP39A1 residues 106-469, followed by a stop.

Junction zoom: aligned by the pinned exon/CDS map

EIF3K 151-186	QHIDRWLLAEMLGDLSDSQLKVWMSKYGWSADESGQ
Fusion 151-186	QHIDRWLLAEMLGDLSASIPKNVFLALHEKLYIMLK
CYP39A1 89-124	KKVDFELAVQNIVYRTASIPKNVFLALHEKLYIMLK

The mutant highlight marks junction codon 167. The novel feature is the joined protein context; the CYP39A1-derived suffix is reference sequence, not a novel frameshift tail. Other reported fusion isoforms remain possible.

Exact original RNA support across the entire coding path



Assembly track: all 1,596 coding bases RNA-witnessed; 0 reference-filled bases.

Each bar counts distinct original fragment IDs with an exact read covering the whole codon. It is not raw BAM depth. The RNA library and ONT library remain separate. Six of 21 ONT candidate reads contain the exact 20+20-base junction; nine span coding start to stop in local collinear alignments. Four satisfy both checks.

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Complete protein aligned to its two reference contributors

D = retained EIF3K reference; F = fusion candidate; A = retained CYP39A1 reference. Dashes mean the contributor is absent at that fusion position, not a deletion from the other gene. The fusion row distinguishes the CYP39A1 suffix and split junction codon. Coordinates at left refer to the fusion protein. The terminal * is the observed stop.

Fusion amino acids 1-60

```
D  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
F  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
A  ------------------------------------------------------------
```

Fusion amino acids 61-120

```
D  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
F  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
A  ------------------------------------------------------------
```

Fusion amino acids 121-180

```
D  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLS--------------
F  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSASIPKNVFLALHEK
A  ---------------------------------------------ASIPKNVFLALHEK
```

Fusion amino acids 181-240

```
D  ------------------------------------------------------------
F  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
A  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
```

Fusion amino acids 241-300

```
D  ------------------------------------------------------------
F  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
A  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
```

Fusion amino acids 301-360

```
D  ------------------------------------------------------------
F  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
A  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
```

Fusion amino acids 361-420

```
D  ------------------------------------------------------------
F  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
A  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
```

Fusion amino acids 421-480

```
D  ------------------------------------------------------------
F  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
A  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
```

Fusion amino acids 481-531

```
D  -------------------------------------------------
F  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
A  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
```

Terminal stop: * | Annotated EIF3K start and CYP39A1 stop retained.

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Full reference proteins and provenance of amino-acid changes

EIF3K / ENST00000248342

1-60	MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
61-120	FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
121-180	LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSDSQLKVWMSKYGWS
181-218	ADESGQIFICSQEESIKPKNIVEKIDFDSVSSIMASSQ

CYP39A1 / ENST00000275016

1-60	MELISPTVIIILGCLALFLLLQRKNLRRPPCIKGWIPWIGVGFEFGKAPLEFIEKARIKY
61-120	GPIFTVFAMGNRMTFVTEEEGINVFLKSKKVDFELAVQNIVYRTASIPKNVFLALHEKLY
121-180	IMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKSLF
181-240	STNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKSAK
241-300	DNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKAIM
301-360	EGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYIIP
361-420	SGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARWFA
421-469	LLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI

Highlighted reference residues contribute to the fusion. Dimmed residues are not retained. The fusion codon at residue 167 contains bases from both partners and encodes CYP39A1 alanine 105; it is not counted as a complete EIF3K-derived codon.

Variant provenance

- Focal change
- Germline
- Phased somatic

No additional non-reference residues occur in the displayed RNA sequence relative to its indicated reference contributor(s). No germline or other somatic allele is assigned without genotype and phasing evidence. Missing sequence stays unobserved.

There is no reference-filled region in the determined ORF. The gene junction at coding offset 499 is an RNA-supported splice/fusion boundary, not an assembly-to-reference stitching point. Support falls outside the enriched junction region, as shown by the codon track; no junction count is assigned to every residue.

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All sequencing and reference inputs used in this focused report

Illumina tumor RNA: public INTEGRATE test-data.tar.gz

Original unmapped.bam (185 fusion-enriched records) and accepted_hits.bam (19 mapped records retained for coding tiling). Archive SHA-256: 9a37130e768e424935008f1d0a67050f6e11a242a263ed2f20ba071d94b767cb. The archive is the source identifier; no SRA run accession is asserted for this subset.

Independent ONT direct RNA: SRR23456709 / PRJNA934933

BioSample SAMN33292477; 1,049,452 reads in the source run. The report checks 21 original candidates and 42 local alignment records. This is a local candidate screen, not a genome-wide uniqueness test.

Tumor and matched-normal DNA: INTEGRATE example archive

All original alignments in two GRCh37 flanking intronic windows were retained. 18 tumor cross-gene fragments versus 0 normal. This supports a rearrangement, without resolving its exact DNA breakpoint.

Exon/cDNA and coding references

Pinned GRCh37-lite archive models: EIF3K ENST00000248342 and CYP39A1 ENST00000275016. Ensembl 87 CDS offsets transfer only after exact whole-transcript sequence identity. ONT placements use GRCh38 and separately pinned Ensembl 87 exon coordinates; genomic coordinates from the two assemblies are never mixed.

Derived files, checksums and broader report source index

The adjacent HCC1395_EIF3K_CYP39A1_fusion.json pins all eight BAM/FASTA/model inputs and contains the complete protein, reference contributors and per-codon support. SEQUENCING_SOURCES.md lists every library, DNA source and reference used in the Sid/HCC1395/COLO829 report set.

The first page highlights strong junction support. It does not assign 149 fragments to every amino acid or claim that 149 reads span the entire protein. Complete coding sequence support is the separately displayed overlap tiling, corroborated by the source-separated ONT checks.